

Lab: Sets, Variables, and Graphs -- The Mathematical Language of Systems

Objective

Build mathematical representations of systems using **set notation**, **variables**, and **graphs**, connecting these tools to the formal definition of Markov blankets and system boundaries.

Prerequisites

- Completed the Everyday Life section on Systems
- Basic algebra (variables, equations)
- Familiarity with set notation (optional -- will be introduced)

Part 1: Set Notation for Systems

Goal: Represent a system's components using formal set notation.

A system consists of internal states, sensory states, active states, and external states. Define these as sets:

1. Let $I = \{i_1, i_2, i_3\}$ be internal states. Give a concrete example (e.g., for a thermostat: $I = \{\text{target temperature, current reading, error signal}\}$).
2. Let $S = \{s_1, s_2\}$ be sensory states. What are the sensory inputs for your example?
3. Let $A = \{a_1, a_2\}$ be active states. What actions can the system take?
4. Let $E = \{e_1, e_2, e_3, e_4\}$ be external states. What is outside the system?
5. Define the Markov blanket as $B = S \cup A$. Write it out for your example.

Write your response here

Part 2: Variables and State Spaces

Goal: Represent system states as variables with ranges.

For a room thermostat system:

1. Let x = current room temperature. What is the range? (e.g., x is in $[0, 50]$ degrees Celsius.)
2. Let s = sensor reading. Is s continuous or discrete?
3. Let a = heater setting. What values can a take? (e.g., {off, low, medium, high}.)
4. Write the state space as the Cartesian product of all variables.
5. How many possible states exist if temperature is discretized into 1-degree bins?

Write your response here

Part 3: Graphs and Networks

Goal: Represent a system as a graph showing dependencies between variables.

Draw or describe a graph where: - Each variable (x , s , a , e) is a node. - An edge from node A to node B means "A influences B."

For the thermostat example: 1. Does the external temperature (e) influence the sensor reading (s)? Draw the edge. 2. Does the sensor reading (s) influence the heater action (a)? 3. Does the heater action (a) influence the room temperature (x)? 4. Does the room temperature (x) influence the sensor reading (s)?

You have drawn a **causal graph**. Identify the cycle (feedback loop) in your graph.

Write your response here

Part 4: The Markov Blanket in Graph Form

Goal: Identify the Markov blanket on your graph.

On your graph from Part 3: 1. Highlight the internal state nodes. 2. Highlight the blanket nodes (sensory + active states). 3. Highlight the external state nodes. 4. Verify that the internal states are only connected to external states *through* blanket states (never directly). This is the Markov property.

Write your response here

Summary Table

Math Tool	What It Represents	Example
Set	A collection of possible values for a variable	$I =$
Variable	A quantity that can take different values	$x =$ room temperature
State Space	All possible combinations of variable values	$X = I \times S \times A \times E$
Graph	A diagram of dependencies between variables	Nodes and directed edges
Markov Blanket	The set of nodes that separate internal from external	$B = S \text{ union } A$